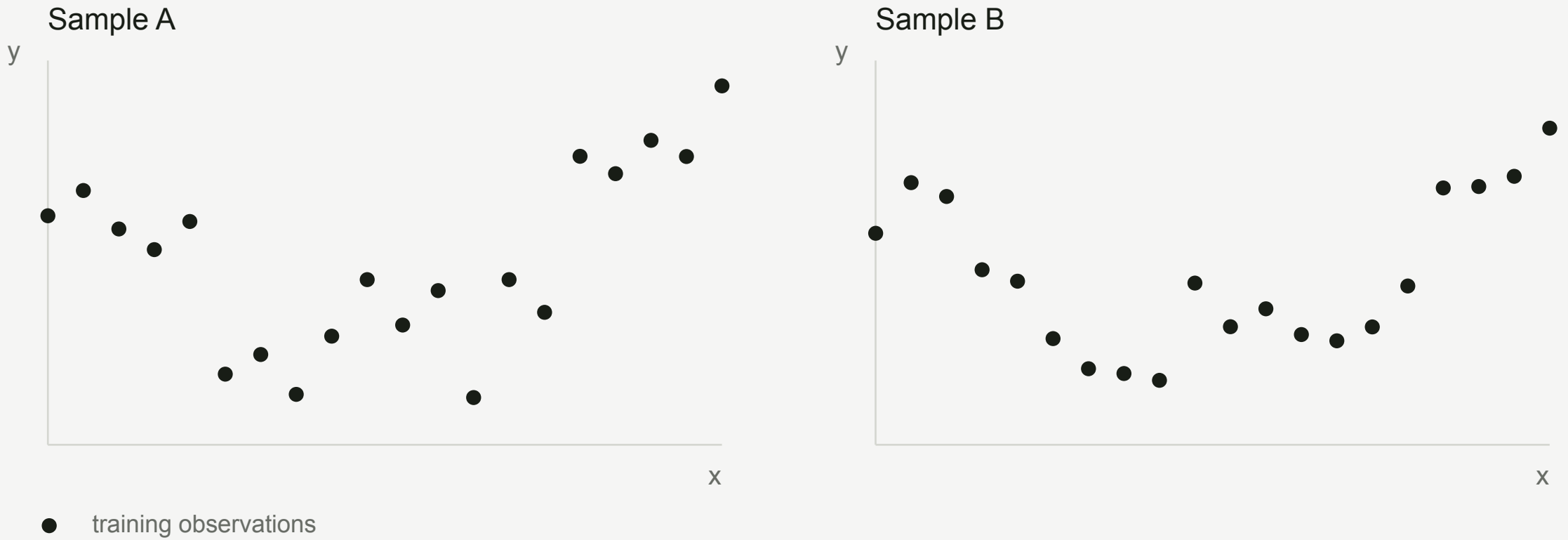
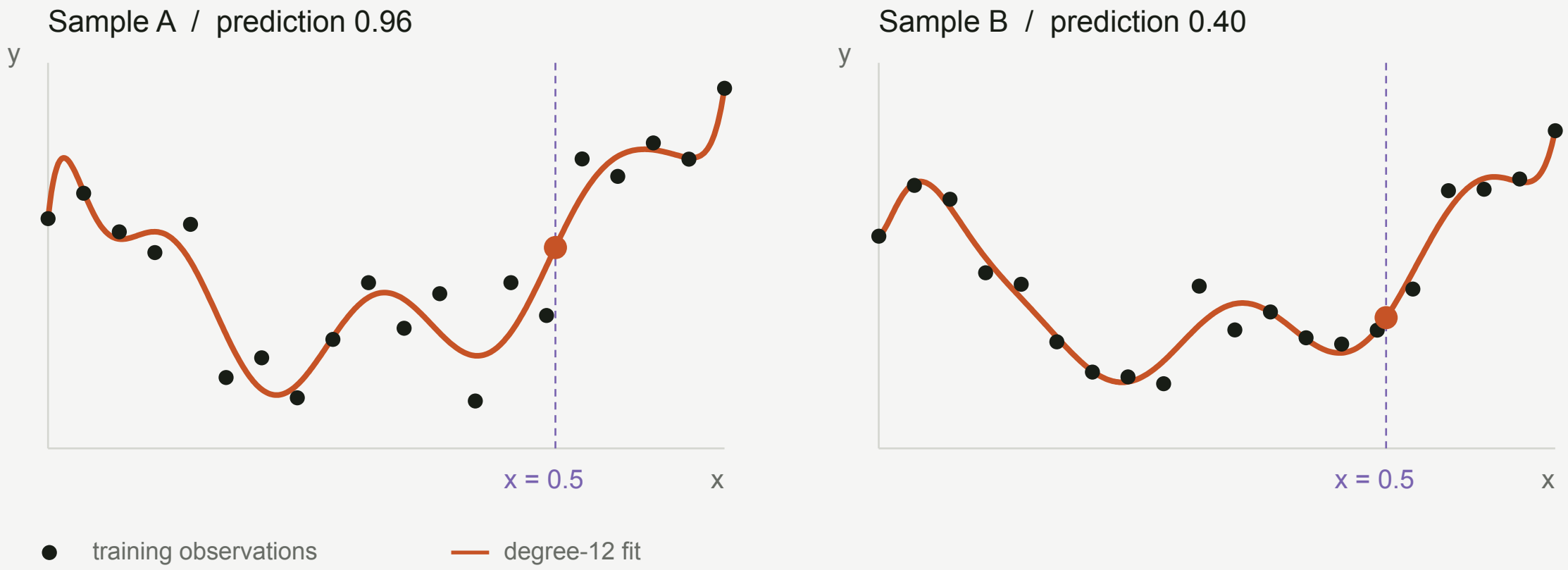
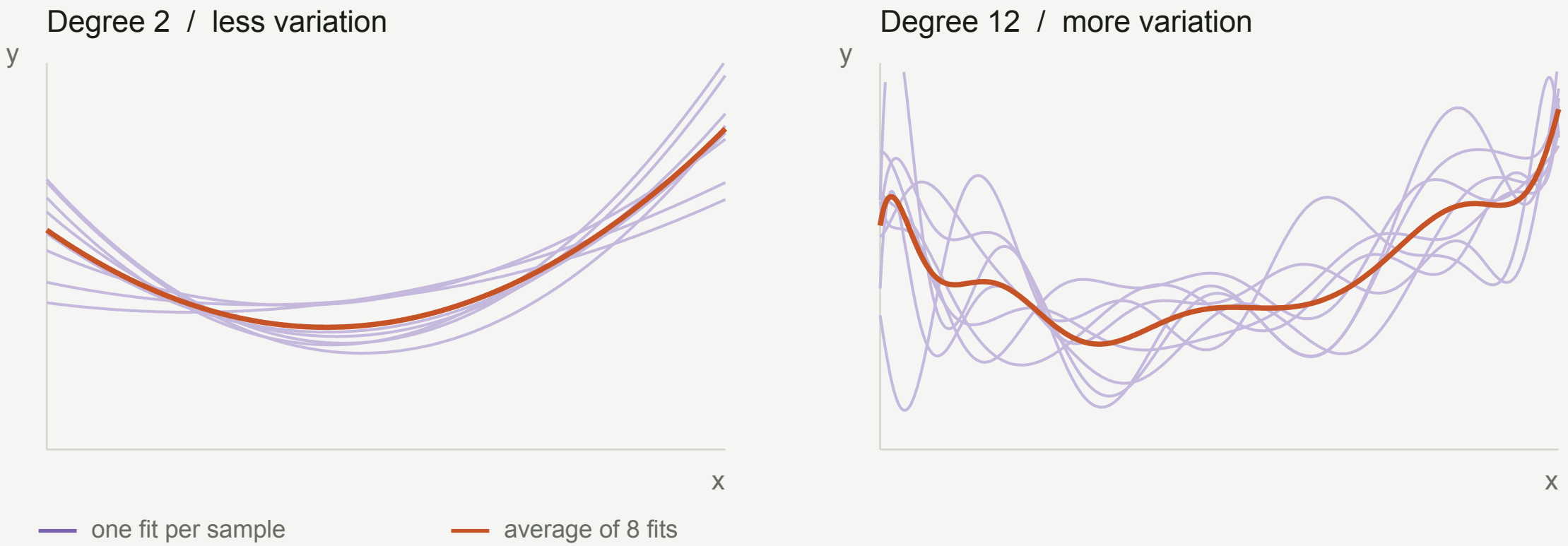
Same problem. Different training samples.



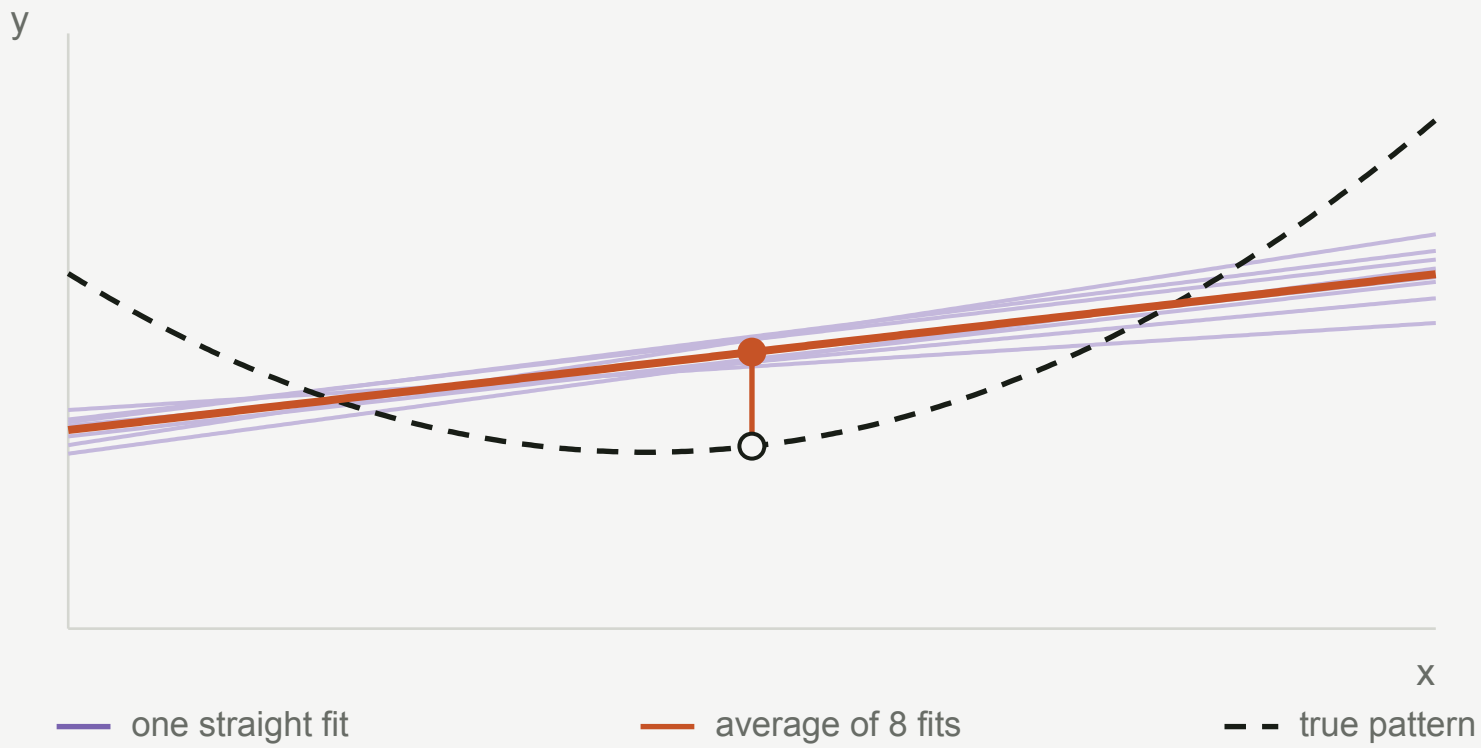
Train again. Does the prediction change?



Variance is sensitivity to the sample.



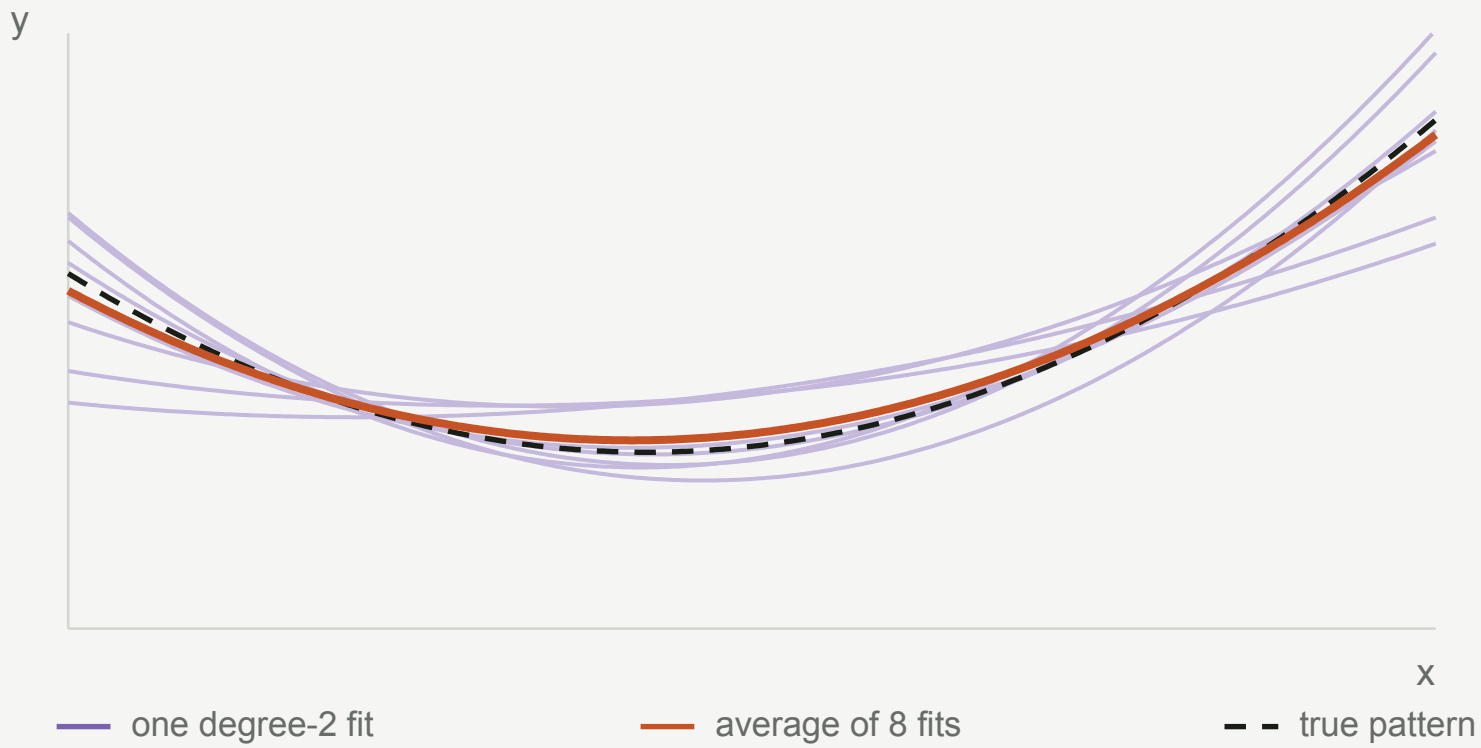
Consistency is not enough.



Bias

Average prediction minus truth.

Close to the truth. Stable across samples.



Low bias.
Low variance.

Three sources of prediction error.

Expected squared error = bias² + variance + noise

BIAS²

Systematic error.

VARIANCE

Sensitivity to the sample.

NOISE

Uncertainty that remains.

Even the true pattern cannot predict every noisy observation.

Learn accurately. Predict consistently.

BIAS

Is the average prediction off?

VARIANCE

How much does retraining change it?